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ManifoldGD: Training-Free Hierarchical Manifold Guidance for Diffusion-Based Dataset Distillation

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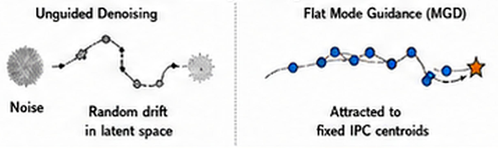
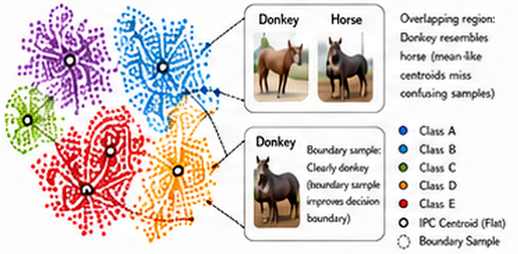
1. Problem & Motivation

From Massive Data to a Compact Distilled Set

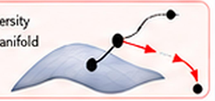
Limitations of Existing Training-Free Methods



Centroid miss boundary samples for better decision boundary



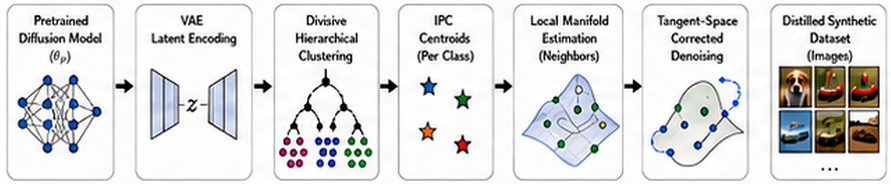
MGD (prior work): Mode guidance improves diversity but lacks manifold consistency → can drift off-manifold and lose fine semantic details.



- Large datasets are expensive to store and train on.
- Flat IPC centroids miss coarse-to-fine intra-class structure.
- Unconstrained mode guidance can *drift off-manifold*, hurting *fidelity and downstream performance*.

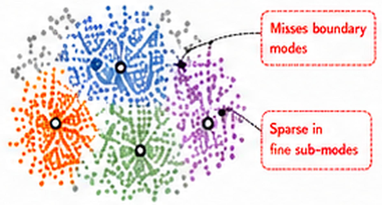
2. Method: Hierarchical IPC Selection + Manifold Guidance

A. Overview



B. Hierarchical IPC Selection

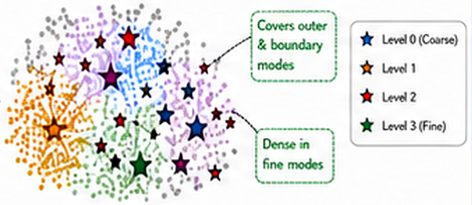
Prior: Flat Centroid Selection (Limited Coverage)



Only a few centroids → misses coarse-to-fine structure

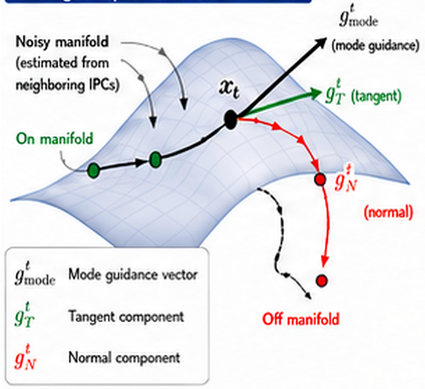
vs.

Ours: Divisive Level-wise Selection (Better Coverage)



Coarse-to-fine IPCs capture center, boundary & sub-modes

C. Tangent-Space Manifold Correction



Score decomposition

$$\nabla_{x_t} \log p_t(x_t \mid c) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_t(c \mid x_t)$$

Marginal Denoising (unconditional score) — Mode Guidance (conditional score)

Manifold guidance modifies the mode-guidance term:

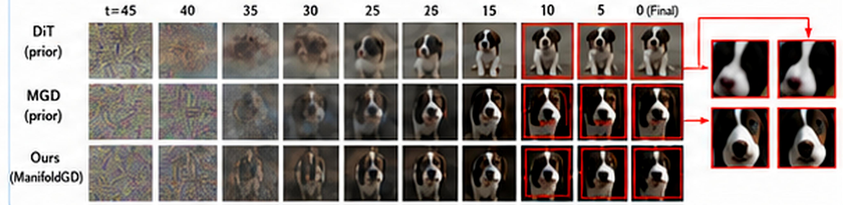
$$g^t_{\text{manifold}}(x_t; c) = g^t_{\text{mode}}(x_t; c) - \mathrm{P}_{N_t}\, g^t_{\text{mode}}(x_t; c)$$

Remove normal component to keep update manifold-consistent

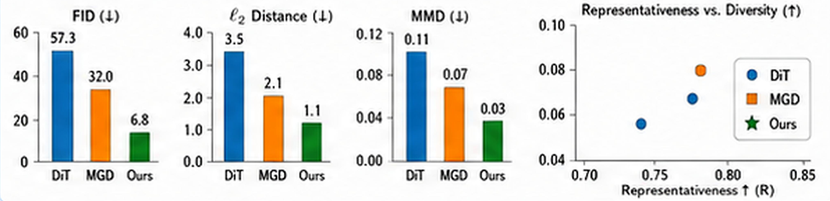
- Divisive clustering provides coarse-to-fine IPC centroids.
- *Local neighbors* define a noisy manifold at each timestep.
- *Tangent projection* preserves semantic attraction while *removing off-manifold drift*.

3. Results & Analysis

1) Qualitative Evolution of Generated Samples across Denoising Timesteps (ImageNette (Dog), IPC=20)



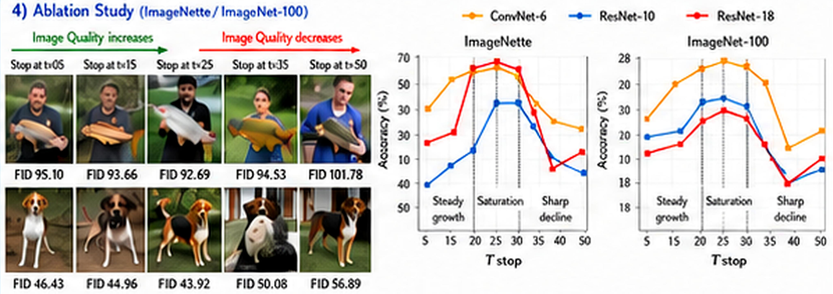
2) Quantitative Metrics on ImageNette-100 (IPC=20)



3) Hard-Label Accuracy from Table 1 (ResNetAP-10)

Dataset	IPC	DiT*	MGD*	ManifoldGD
ImageNette	10	59.1 ± 0.7	61.9 ± 0.6	**64.1 ± 1.2** (+2.2)
	20	64.8 ± 1.2	66.5 ± 0.5	**69.7 ± 0.4** (+3.2)
	30	73.3 ± 0.9	77.5 ± 1.1	**78.4 ± 1.0** (+3.4)
ImageNet-100	10	23.2 ± 0.3	26.1 ± 0.6	**27.6 ± 0.5** (+1.5)
	20	28.4 ± 0.4	33.2 ± 0.5	**35.3 ± 0.5** (+2.1)

4) Ablation Study (ImageNette / ImageNet-100)



- ManifoldGD improves FID, ℓ_2, MMD, representativeness, diversity, and $Acc_{S\to D}$.
- Offers better geometric/manifold fidelity than prior works (MGD, DiT).
- First geometry-aware, training-free dataset distillation framework for diffusion models.